Basemental FG 250: User Manual

Version 1.3.0 · two-channel fusion guitar amp plug-in with strobe tuner and FET Stack compressor



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1. Install

Download the installer for your computer from the Releases page.

Mac (Logic Pro, GarageBand, MainStage, Ableton, Cubase…)

1. Quit your DAW.
2. Double-click **BasementalFG250-1.3.0-macOS-Installer.pkg**.
3. If macOS says it can't verify the developer: click **Done**, open **System Settings › Privacy & Security**, scroll down, click **Open Anyway** next to the Basemental message, then run the installer again.
4. Click through the installer. Keep both **Audio Unit** (Logic, GarageBand, MainStage) and **VST3** (other DAWs) ticked.

Works on Apple Silicon and Intel Macs. Files go to `/Library/Audio/Plug-Ins/Components` (AU) and `/Library/Audio/Plug-Ins/VST3`.

Windows

1. Run **BasementalFG250-1.3.0-Windows-Setup.exe**.
2. If SmartScreen appears, click **More info › Run anyway**.
3. The VST3 goes to `C:\Program Files\Common Files\VST3`; a standalone app is added to the Start menu. Uninstall from **Settings › Apps**.

Linux

Unpack **BasementalFG250-1.3.0-Linux.tar.gz** and run `./install.sh` (VST3 to `~/.vst3`, LV2 to `~/.lv2`). `./install.sh --uninstall` removes it.

REAPER (no install)

Copy **BasementalFG250-REAPER.jsfx** into REAPER's `Effects` folder (**Options › Show REAPER resource path**). The REAPER version has the amp, cabinets and guitar voices, but not the tuner or FET Stack.

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2. Quick start in Logic Pro

1. Create an **Audio** track and set its input to the interface input your guitar is plugged into. Turn on **Input Monitoring** (the "I" button).
2. Click an **Audio FX** slot › **Audio Units › Basemental › Basemental FG 250**.
3. Play and set the **INPUT** knob (bottom right) so the **IN** VU meter swings around **0** on your hardest notes. The small red light in the meter's corner must stay off.
4. Choose a sound from the **preset menu** (top right), or start from **Init**.
5. Click **TUNER** to tune, **COMP** to open the compressor.

> Not listed in Logic? Quit and reopen Logic. Still missing: **Logic Pro › Settings › Plug-in Manager**, select Basemental FG 250, **Reset & Rescan Selection**.

Signal flow

```
Guitar → Input → Guitar Voice → FET Stack (if on) → CLEAN or LEAD channel
       → Reverb → Power amp → Cabinet + Mic (or your IR) → Master → Logic
```

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3. The amp

Channels

Control	What it does
LEAD / CLEAN button	Switches channels (red light = Lead, green = Clean). The switch crossfades, so it never clicks.
GAIN	Clean: from spotless to edge-of-breakup. Lead: from light crunch to singing sustain. Lead Gain starts at **0.51**; double-click the knob to return there.
LOW / MID / HIGH	Tone controls after the gain stage. **5 = flat**. Low and High shelve ±15 dB, Mid is a broad ±12 dB.
LEVEL	Channel volume into the power amp. Above about 7 it starts to push the power stage, like a cranked amp.

Control	What it does
REVERB	Shared reverb amount. The chrome **footswitch** under it turns reverb on/off (red light = on).
POWER	Rocker switch. Off fades the amp to silence; the blue **ON** lamp shows it's on. The tuner keeps working when power is off.

Each channel remembers its own settings, so the button gives you a ready clean/lead pair.

Bottom panel

Control	What it does
GUITAR VOICE	Gentle EQ that nudges your DI toward a pickup or body character: FG1 Neck/Bridge, FG2, single-coil, humbucker, hollowbody, **Semi-Hollow** (default), and more. Choose **Off** to hear your own guitar exactly.
CABINET	Eight built-in cabinets, from 1×8 to 4×12. Used when Output is *Built-in Cab*.
MIC	On-Axis (brightest), Off-Axis (darker, smoother), Close (more low end).
OUTPUT	*Built-in Cab*; *User IR* (your own cabinet impulse, load it with **Load IR...**); or *Pre Out*, which skips the power amp and cabinet so you can use your own.
-6 dB	Pads the Pre Out by 6 dB.
INPUT	Trim before everything (-12 to +12 dB). Set this with the IN meter.
MASTER	Final output level (-24 to +12 dB).
IN / OUT VU meters	0 VU = -18 dBFS. Aim for the needle around 0 on loud notes. The red corner light latches if the signal hits 0 dBFS (clipping); click the meter to reset it.

Double-click any knob to reset it. Drag up/down to turn; hover to see its value.

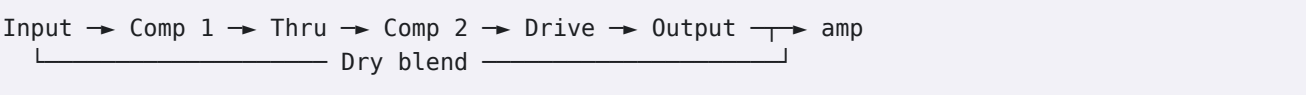
Amp presets (top-right menu)

Init, plus eight Gambale-style tones: Glassy Clean, Jazz Neck Clean, Edge of Breakup, Sweep Lead, Fat Round Lead (neck), Singing Sustain, Fusion Crunch Rhythm, Ballad Lead (wet). Amp presets change the amp only; they don't touch the FET Stack. They're interpretations of tones Frank Gambale has described, not his actual settings.

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4. FET Stack compressor

Click **COMP** in the top bar. The FET Stack works like a compressor pedal at the start of your board: it sits **before the amp**. When it's **off**, your guitar goes **straight to the amp**, completely untouched.



Controls

Control	What it does
ON (footswitch)	Turns the FET Stack on/off. Red light = on.
INPUT	How hard you hit the compressors (-24 to +24 dB). The threshold is fixed, so more Input = more compression.
ATTACK	20 µs (instant: flattens the pick) to 50 ms (lets the pick attack through).
RELEASE	50 ms (snaps back, can pump) to 1.1 s (smooth, long sustain).
AUTO	Automatic release: quick after short peaks, slow during sustained playing, so it levels without "breathing". Overrides RELEASE.
KNEE	0 dB = hard knee (grabs abruptly), 12 dB = very soft (eases in).
THRU	Level from Comp 1 into Comp 2: how hard the second compressor works.
DRIVE	Warm Class-A-style saturation with even and odd harmonics. Adds harmonics, not volume; never hard-clips.
DRY	Blends your uncompressed signal back in (parallel compression). 0 % = all compressed.
OUTPUT	Make-up level after compression (-24 to +24 dB).
RATIO	2, 3, 4, 6, 8, 12, 20 : 1, or **ALL**: "all buttons in" mode, where the compressor over-compresses (hit harder and it gets *quieter*), with an aggressive, distorted character.
STACKED	On: Comp 1 feeds Comp 2. Off: Comp 1 only.
LINE DRIVE	Comp 1 stops compressing and becomes an overdriven preamp: INPUT sets how hard it's driven, DRIVE sets the character.
Gain-reduction meters	How many dB each compressor is turning the signal down.

FET Stack presets (menu at the top of the COMP panel)

FET Stack presets **change only the compressor**, never the amp, and switch the FET Stack on. Each one's Output is set so it plays at the same loudness as the compressor switched off, so you can A/B fairly with the ON footswitch.

Preset	Compression	Good for
Gentle Always-On	~3 dB	Light levelling you leave on all the time
Jazz Smooth Clean	~5 dB	Smooth jazz cleans that keep their dynamics
Clean Arpeggio Shimmer	~6 dB	Even arpeggios and chord melodies
Country Snap	~8 dB	Chicken pickin', snappy single notes
Funk Rhythm Squash	~12 dB	Tight, percussive funk rhythm
Parallel Punch	~15 dB	Heavy squash blended 50/50 with your dry tone
Sweep Picking Evener	~8 dB	Even note volume across sweeps and fast picking
Fusion Legato Sustain	~10 dB	Smooth legato lines

Preset	Compression	Good for
Singing Lead Sustain	~16 dB	Long, singing lead notes
Stacked British Mode (Zeppelin-style DI)	~12 dB	Driven preamp into all-buttons compression: raw, pumping rock
Sustain for Days (Gilmour-style)	~14 dB	Soaring solos with endless sustain
16th-Note Chop (Chic-style funk)	~10 dB	Uniform 16th-note funk strumming
LA Studio Clean (80s session snap)	~8 dB	Punchy studio cleans: the pick passes, the tail is held
Wall-to-Wall Acoustic (modern pop)	~6 dB	Level, upfront strumming (acoustic or clean electric)
Front-End Anchor (djent tightener)	~8 dB	Tightening low-tuned riffs before the Lead channel

The style presets are inspired by commonly described approaches; they aren't the artists' or engineers' actual settings and aren't endorsed by them.

Tips

- Start from a preset, then adjust **INPUT** for more or less compression and **OUTPUT** to match the level.
- Compression depends on how hot your guitar arrives: with a quiet DI, raise the FET Stack INPUT; with hot pickups, lower it.
- For more sustain without more squash, add a little **DRIVE**.
- Blend in **DRY** (20–40 %) if a sound feels too flat.

5. Strobe tuner

Click **TUNER** in the top bar. The tuner reads your guitar **before** the amp. With **Mute** ticked, the output is silent while you tune.

Reading it

- The large note and its octave, the deviation in **cents to 0.1**, and the frequency to 0.01 Hz.
- **Strobe bands**: they stand still when you're in tune, drift **right when sharp** and **left when flat**, faster the further off you are. The band for the note's octave shows the crisp pattern; higher bands magnify small errors.
- **FLAT / IN TUNE / SHARP** lights: green within ±0.5 cent.

Accurate to better than 0.1 cent (measured worst case 0.06 cent). It locks in about 0.1–0.2 s on guitar notes and up to 0.4 s on low bass notes.

Settings

Setting	What it does
A =	Reference pitch, 415–466 Hz in 0.1 Hz steps (default 440).
OFFSET	Shifts every target by up to ±50 cents in 0.1-cent steps.

Setting	What it does
AUTO / MANUAL	Auto finds the nearest note. Manual tunes to the note you pick with the 12 note buttons (useful for harmonics or big detunes, up to ±250 cents).
KEY	Shows notes for C (concert), B♭, F or E♭ instruments.
Temperament + root	Equal (default), Just, Pythagorean, 1/4-Comma Meantone, Werckmeister III, Kirnberger III, Vallotti, Young II.
String guide	Lights the string you're tuning: Chromatic, Standard, **Gambale (A2 D3 G3 C4 E3 A3)**, Half Step Down, Drop D, 7-String.
Mute	Silences the output while the tuner is open.

Tips: pluck one string at a time, let the note ring, and tune *up* to the note. Very loud mains hum from single-coils can shake the last decimal on low strings; move or turn slightly to reduce the hum.

6. Troubleshooting

Problem	Fix
Not in Logic's plug-in list	Quit and reopen Logic, then **Settings › Plug-in Manager › Reset & Rescan Selection**.
Mac won't open the installer	**System Settings › Privacy & Security › Open Anyway**, then run it again.
No sound	Check **POWER** is on (blue lamp), the tuner is closed (it mutes), and Logic's Input Monitoring is on.
Sound is distorted when it shouldn't be	Lower **INPUT** until the IN meter's red light stays off; check LEAD/CLEAN.
Too quiet / too loud	Use **MASTER**, or OUTPUT on the FET Stack.
Compressor does nothing	Turn up FET Stack **INPUT**; check the ON light.
Compressor too strong	Lower FET Stack **INPUT**, raise **DRY**, or try *Gentle Always-On*.
Tuner jumps between notes	Mute the other strings and pluck firmly; use **MANUAL** for harmonics.
Latency when playing live	Lower Logic's I/O buffer size (**Settings › Audio**). The plug-in adds no latency of its own.

7. Specifications

- Formats: AU (macOS), VST3 (macOS, Windows, Linux), LV2 (Linux), Standalone, REAPER JSFX; AAX can be built with Avid's SDK
- Mono or stereo input, stereo output (the reverb is stereo)
- Sample rates: 44.1–192 kHz; zero added latency
- Internal 4× oversampling on the saturation stages
- Tuner: strobe (phase-tracking) measurement, better than 0.1 cent
- Source code: https://github.com/cmacjam/Basemental-Frank-Gambale

Basemental FG 250 is an independent project. DV Mark, Kiesel and the other product and artist names mentioned belong to their owners and are used only to describe the sounds the plug-in is modelled on.